

Faculty of Computing

Object Oriented Programming Lab

Lab Report 1

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Lab Task 1 — Calculate Average

Create a function float calculate Average(int arr[], int size) that receives an integer array and its size and returns the average of its elements. In main(), ask the user for the number of elements, validate that it is positive, input the elements, call the function, and display the average.

```
1  #include <iostream>
2  using namespace std;
3
4  float calculateAverage(int arr[], int size) {
5      int sum = 0;
6
7      for (int i = 0; i < size; i++) {
8          sum += arr[i];
9      }
10
11     return (float)sum / size;
12 }
13
14 int main() {
15     int size;
16
17     cout << "Enter number of elements: ";
18     cin >> size;
19
20
21     if (size <= 0) {
22         cout << "Invalid size! Number of
   elements must be positive.";
23         return 0;
24     }
25
26     int arr[size];
27
28
29     cout << "Enter " << size << " elements:\n";
```

```
30      for (int i = 0; i < size; i++) {
31          cin >> arr[i];
32      }
33
34
35      float average = calculateAverage(arr,
    size);
36
37      cout << "Average = " << average << endl;
38
39      return 0;
40  }
```

Output

```
Enter number of elements: 3
Enter 3 elements:
23
45
32
Average = 33.3333

[Program finished]
```

Lab Task 2 — Student Records Define a Student struct with string name, int roll Number, and float marks. Create an array of three Student struct, input the information for each student, and display all records.

```
1  #include <iostream>
2  using namespace std;
3
4  struct Student {
5      string name;
6      int rollNumber;
7      float marks;
8  };
9
10 int main() {
11     Student students[3];
12
13     |
14     for (int i = 0; i < 3; i++) {
15         cout << "Enter details for student " << i
   + 1 << ":\n";
16
17         cout << "Name: ";
18         cin >> students[i].name;
19
20         cout << "Roll Number: ";
21         cin >> students[i].rollNumber;
22
23         cout << "Marks: ";
24         cin >> students[i].marks;
25     }
26
27
28     cout << "-- Student Records --";
29
```

```
28      cout << "-- Student Records --";
29
30      for (int i = 0; i < 3; i++) {
31          cout << "Student " << i + 1 << endl;
32          cout << "Name: " << students[i].name
    << endl;
33          cout << "Roll Number: " << students[i].
    rollNumber << endl;
34          cout << "Marks: " << students[i].marks
    << endl;
35          cout << endl;
36      }
37
38      return 0;
39  }
40
```

Output:

```
Enter details for student 1:
Name: lubna
Roll Number: 12
Marks: 46
Enter details for student 2:
Name: romi
Roll Number: 35
Marks: 79
Enter details for student 3:
Name: arslan
Roll Number: 24
Marks: 89
-- Student Records --Student 1
Name: lubna
Roll Number: 12
Marks: 46

Student 2
Name: romi
Roll Number: 35
Marks: 79

Student 3
Name: arslan
Roll Number: 24
Marks: 89


[Program finished]
```

Lab Task 3 — Check Number

Write a C++ program that asks the user to enter a number and checks whether the number is positive, negative, or zero. Use appropriate conditional statements to make the decision and display the result.

```cpp
#include <iostream>
using namespace std;


int main() {
    int number;


    cout << "Enter a number: ";
    cin >> number;


    if (number > 0) {
        cout << "The number is positive." << endl;
    }
    else if (number < 0) {
        cout << "The number is negative." << endl;
    }
    else {
        cout << "The number is zero." << endl;
    }


    return 0;
}
```

Output:



Lab Task 4 — Find Maximum

Write a C++ program that stores five integers in an array and creates a function int find Maximum(int arr[], int size) to find and return the largest element in the array. In main(), input the elements, call the function, and display the largest number.

```cpp
#include <iostream>
using namespace std;


int findMaximum(int arr[], int size) {
    int maximum = arr[0];


    for (int i = 1; i < size; i++) {
        if (arr[i] > maximum) {
            maximum = arr[i];
        }
    }


    return maximum;
}


int main() {
    int arr[5];


    cout << "Enter 5 integers:" << endl;

```

```
24      cout << "Enter 5 integers:" << endl;
25
26
27      for (int i = 0; i < 5; i++) {
28          cin >> arr[i];
29      }
30
31
32      int largest = findMaximum(arr, 5);
33
34
35      cout << "The largest number is: " <<
    largest << endl;
36
37
38      return 0;
39  }
```

Output


Enter 5 integers:
65
23
23
44
56
The largest number is: 65
[Program finished]